

DishDive Application: Menu Review Analysis

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Abstract

In the modern dining experience, diners face a time-consuming and inefficient process of sifting through scattered online reviews across multiple platforms to select a meal. This project addresses this inefficiency by designing and developing DishDive, an AI-powered mobile application that analyzes online customer feedback to recommend top-rated dishes in Bangkok.

The primary objective was to create a centralized system that extracts, processes, and summarizes social media comments to provide actionable, dish-level insights. The methodology involved implementing a comprehensive system architecture, beginning with a web crawling module to gather review data from platforms like Facebook and X. This data was then processed through a pipeline utilizing a Large Language Model (LLM) to perform sentiment analysis and keyword extraction.

The result of this project is a functional DishDive application that provides users with a searchable restaurant database and personalized dish recommendations. The system successfully transforms unstructured, scattered opinions into summarized, data-driven insights, significantly reducing the time users spend on manual searches and enhancing their ability to make informed dining decisions.

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Chapter 1

Introduction

1.1 Background

In the modern dining experience, selecting a meal is often time-consuming and frustrating. Customers rely on scattered online reviews and social media discussions, which makes it difficult to identify the best dishes. The lack of a centralized and structured system for aggregating and analyzing this information leads to inefficiencies and uncertainty when choosing what to order. Users must manually sift through multiple sources to determine popular menu items, creating information overload and suboptimal choices. Restaurants and food enthusiasts therefore lack a streamlined solution that compiles feedback into meaningful insights.

1.2 Objectives

This project aims to develop DishDive, an AI-powered system that analyzes online customer feedback to recommend top-rated dishes in Bangkok. It will extract and process social media comments, perform sentiment analysis, and summarize insights. The system provides a searchable restaurant database, an interactive summary map, and personalized recommendations based on user preferences. A mobile application (iOS/Android) serves as the primary user-friendly interface, ensuring a seamless experience.

1.3 Scope

extbfFunctions and Features:

- Analyze social media comments and reviews to surface dish-level insights.
- Provide a restaurant database with search, filtering, and personalization.
- Deliver recommendations based on user preferences and blacklists.
- Offer mobile access (iOS/Android) as the primary interaction channel.

extbfUsers:

- Diners seeking the best dishes in Bangkok.
- Restaurants interested in structured feedback to improve menus.

extbfSupported Platforms:

- Mobile application (iOS/Android) as primary; backend services and database as supporting components.

1.4 Expected Benefits

DishDive is expected to reduce the time users spend manually searching for reviews by transforming unstructured, scattered opinions into summarized, data-driven insights. Users gain confident, informed dining decisions; restaurants gain actionable feedback to refine their offerings. Personalized and up-to-date recommendations improve user satisfaction, while structured analytics help restaurants optimize menus.

Figure 1 placeholder

Figure 1.1: Figure 1: Example Placeholder

Chapter 2

Feasibility Study

2.1 Introduction

The feasibility section evaluates the practicality of implementing the proposed system, which extracts and processes social media comments to generate restaurant and dish recommendations. Existing sentiment analysis tools and recommendation systems provide restaurant-level insights but often fail to analyze dish-specific feedback, leaving diners uncertain about the best meal choices. This section assesses the technical, operational, and financial feasibility of delivering precise, dish-level insights in real-world use.

2.2 Problem Statement

Selecting a meal is inefficient because customer reviews are scattered across Facebook, X, Instagram, and other platforms. Users manually search multiple sources to identify recommended dishes, creating information overload and inconsistent signals. Sentiment and keyword analysis are not readily available, making it hard to judge quality and popularity of specific menu items. Restaurants also lack structured insights, limiting their ability to improve offerings. Without a centralized, AI-powered system to summarize feedback, both users and restaurants face frustration and missed opportunities.

2.3 Related Research and Projects

Prior work in AI-driven sentiment analysis and recommendation systems has improved restaurant-level ratings, but gaps remain. DishDive targets dish-specific recommendations, integrates AI-driven sentiment analysis and keyword extraction, and leverages real-time social media data to surface trending dishes. User preference features personalize suggestions, making the system more interactive and tailored to individual tastes.

2.4 Requirement Specifications

Based on initial evaluation, the implementation must satisfy:

- Functional: ingest social data, perform sentiment and keyword extraction, store structured results, and deliver personalized recommendations.
- Non-functional: maintain reliability, responsiveness, and up-to-date data freshness while scaling to ongoing social feeds.

2.5 Implementation Plan

The project is estimated at roughly 5–6 months, divided into phases such as research and wireframing, database design, backend preparation, demo, web crawling, LLM processing, preference algorithm, frontend display, integration and testing, and finalization. Each phase incrementally delivers working components toward a containerized, deployable DishDive application.

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Figure 2.1: Figure 2: Example Placeholder

Chapter 3

System Analysis and Design

3.1 Analysis of the Existing System

The current approach to discovering restaurant dishes relies on manual searches across social platforms, food blogs, and review sites. Users face unstructured information and inefficiencies when identifying the best dishes, and there is no automated, structured recommendation workflow.

3.1.1 Current System Workflow

- Google Reviews: integrated into the Google ecosystem to manage and display user-generated reviews across services.
- Wongnai (LINE MAN Wongnai): Thai platform for restaurant reviews, food delivery, and lifestyle content, serving users and merchants.
- Tabelog: Japanese review platform with heavy user-generated content and rankings.

3.1.2 Challenges in the Existing System

- Lack of personalization; dish-level sentiment is rarely surfaced.
- Limited dish-level insights (e.g., Tabelog seldom breaks down to individual dishes).
- Time-consuming decision-making due to scattered data and manual review.
- No real-time data processing; insights are not kept current.

3.2 User Requirement Analysis

Requirements gathered from surveys and interviews with diners and review-platform users include:

- Personalized, accurate recommendations tailored to taste, dietary needs, and history.
- Real-time, reliable information aggregated from blogs, review platforms, and social media.
- Intuitive, user-friendly interface with advanced search and filtering (location, cuisine, dietary restrictions).
- Comprehensive dish insights with sentiment trends and key feedback.
- Efficient performance under load; interactive features to save favorites, submit reviews, and refine recommendations.

3.3 New System Design

The DishDive design introduces AI-driven sentiment analysis, real-time data extraction, and structured recommendations to automate and optimize discovery of the best dishes. A backend stack (data collection, LLM processing, database, recommendation engine) supports a mobile-first front end. (Diagrams such as the site map, data flow, activity diagrams, and system architecture belong here once recreated.)

Figure 3 placeholder

Figure 3.1: Figure 3: Example Placeholder

Chapter 4

System Functionality

4.1 System Functionality

The DishDive application addresses decision fatigue in food choice by transforming unstructured textual reviews into actionable, personalized menu insights. Core functions include:

4.1.1 User Management and Personalization

Secure registration and login via JWT, with user-driven personalization such as Favorites that feed the recommendation logic.

4.1.2 Restaurant and Menu Browsing

Search and browse restaurants and menus, including location-based search and detailed menu views with review analyses.

4.1.3 Review Submission and Real-Time AI Extraction

Users submit reviews in-app; the API immediately triggers AI extraction to pull attributes, sentiment, and topic clusters so new feedback is instantly incorporated.

4.1.4 Menu Review Analysis and Synthesis

Summaries surface flavor profiles, sentiment breakdowns, and recommendation signals derived from batch and real-time reviews instead of raw text streams.

4.2 System Architecture

DishDive uses a modern three-tier architecture for scalable, real-time AI processing.

4.2.1 High-Level Overview

Three layers: Client App (Flutter/Dart UI, state, HTTP), API Server (Go + Fiber, GORM, JWT, real-time LLM calls), and Storage & AI Backbone (PostgreSQL, MinIO, Python batch tools). The runtime flow: Flutter requests → Go API → PostgreSQL or OpenAI LLM services.

4.2.2 Architecture Modules and Stack

extbfClient (Flutter/Dart): Provider for state, Dio for networking, Maps_flutter + geolocator for maps/location.

extbfAPI Server (Go): Fiber v2 for routing, GORM for ORM to PostgreSQL, JWT for auth, native HTTP client for OpenAI calls to perform real-time extraction.

extbfStorage and AI Backbone: PostgreSQL for structured data (users, restaurants, reviews, extracts); MinIO for media assets; Python batch scripts (Pandas, psycopg2, OpenAI client) for offline ingestion/cleaning.

4.2.3 Critical Data Flows

extbfReview Submission and AI Extraction: Flutter sends POST /SubmitReview; Go API stores raw text; calls OpenAI for structured extraction; stores to `review_extracts`; background normalization maps into domain tables and updates recommendation signals.

extbfAuthentication: Login returns JWT; client stores token; subsequent requests include Authorization header; Go API verifies before access.

4.3 Test Plan

Testing focused on end-to-end validation of critical components using manual scenario execution (smoke tests) to verify assessable features F1–F7.

4.3.1 Distinct Project Features Tests

- **F1:** Batch processing of raw review data with LLM, persisting structured data to `review_extracts`.
- **F1.1:** Run web crawler/ingestion to process social comments.
- **F1.2:** Verify sentiment and keyword fields for a sample dish in `review_extracts`.
- **F2:** UI shows summary and top-rated dishes from processed data.
- **F2.1:** Map view displays restaurant locations.
- **F2.2:** Restaurant page shows analyzed menu insights (flavor profiles, service comments).

4.3.2 Support Project Features Tests

- **F3:** Restaurant list populates correctly.
- **F4:** Search by name/location filters list.
- **F5:** Favorites persist and appear in Favorites tab.
- **F6:** Registration and login succeed.
- **F7:** All components (Flutter, Go API, Postgres, MinIO) run in the containerized environment.

4.4 Test Results

All assessable features F1–F7 were executed and marked successful:

- **Distinct Features (F1, F1.1, F1.2, F2, F2.1, F2.2):** Batch processing, sentiment/keyword extraction, summarized dish recommendations, and map/list UI flows all passed.

- **Support Features (F3–F7):** Listing, search, favorites, auth, and full stack deployment validated in the containerized setup.

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Figure 4.1: Figure 4: Example Placeholder

Chapter 5

Summary and Suggestions

5.1 Project Summary and Conclusion

DishDive met its core objective: using AI to convert unstructured restaurant reviews into concise, actionable menu insights for users facing decision fatigue. By synthesizing sentiment and keywords via an LLM, the system delivers summarized recommendations. The functional prototype (Flutter, Go, PostgreSQL) implemented and validated all seven assessable features (F1–F7), showing the viability of AI-driven menu analysis and providing a scalable, maintainable application.

5.2 Problems Encountered and Solutions

Two main technical challenges required adaptation during data acquisition and AI processing.

5.2.1 Web Scraper and Data Acquisition

Automated scraping handled Google Maps reviews but struggled with high-review-count Wongnai shops due to dynamic loading and protections. Solution: dual strategy with automated scraping for shops under 8 reviews and manual-assisted scraping for heavier pages. Roughly 340 shops required manual work over three days to reach the needed data volume.

5.2.2 Large Language Model Selection

A local open-source model (Ollama Qwen) could not produce reliable structured JSON for the complex prompt. Pivoting to GPT-4o mini immediately solved extraction quality, validating the core approach and enabling completion of API and app integration.

5.3 Suggestions for Further Development (DishDive 2.0)

5.3.1 Enhanced Personalization and Recommendation Engine

Move beyond aggregate sentiment to collaborative filtering based on user history and preferences for truly personalized dish recommendations.

5.3.2 Broader Data Acquisition and Scaling

Expand beyond Bangkok: ingest other cities/countries and add live social media feeds (X, Instagram) for real-time trends and comments.

5.3.3 LLM Pipeline Optimization

Adopt a cascading strategy to reduce costs: send simple cleaning/sentiment tasks to a cheaper model, reserve GPT-4o mini for complex extraction and summarization.

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Figure 5.1: Figure 5: Example Placeholder

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